

Michealmas Week 3 & 4

lw703

1 Completed Task

I spend time on understanding the codes of DiKL paper and try to run the codes.

I have tried on the alternative loss function as discussed in last report but it has been shown that the effect of using analytically ground truth score is worse than use a model to fit the score. The main reason should be that when a model fits a score, it is more generalized and provide a stronger robustness.

This is the result I got for the MOG40 dataset - run the DiKL code after I understand it. I read more paper and find a proof for the validity of the DiKL loss:

minimizing DiKL ensures that the neural sampler approximates the target distribution:

$$D(\tilde{p}||q) = 0 \Leftrightarrow D(\tilde{p}||\tilde{q}) = 0, \quad (1)$$

$$\Leftrightarrow \tilde{p} = \tilde{q} \quad (2)$$

$$\Leftrightarrow \mathcal{F}\{k\}\mathcal{F}\{p\} = \mathcal{F}\{k\}\mathcal{F}\{q\}, \quad (3)$$

$$\Leftrightarrow \mathcal{F}\{p\} = \mathcal{F}\{q\}, \quad (4)$$

$$\Leftrightarrow p = q \Leftrightarrow \mathbb{P} = \mathbb{Q}. \quad (5)$$

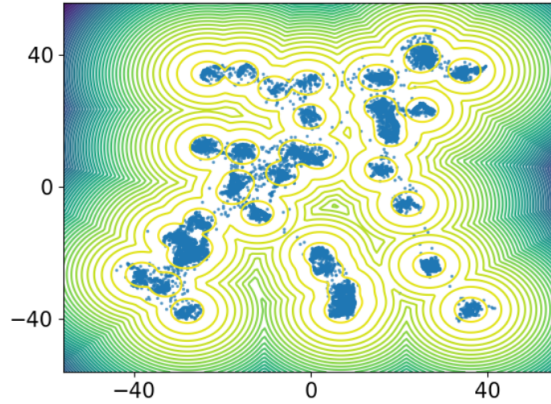


Figure 1: Generation results for DiKL on MOG-40.